

Electrical Wiring

Basics

Cable Types • Conductor Sizing • Installation Methods • Colour Codes

What You Will Learn

- Cable types: T&E, SWA armoured, flexible, screened — construction and selection
- Conductor sizing by current rating, voltage drop, and fault protection (IEC 60364)
- Installation methods: surface clipping, conduit, trunking, and underground
- Wire colour codes (IEC/EU, old UK, USA NEC), connection types, and best practices



Electrical Wiring — Safe, Compliant, Professional

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ELEC-007: Electrical Wiring Basics

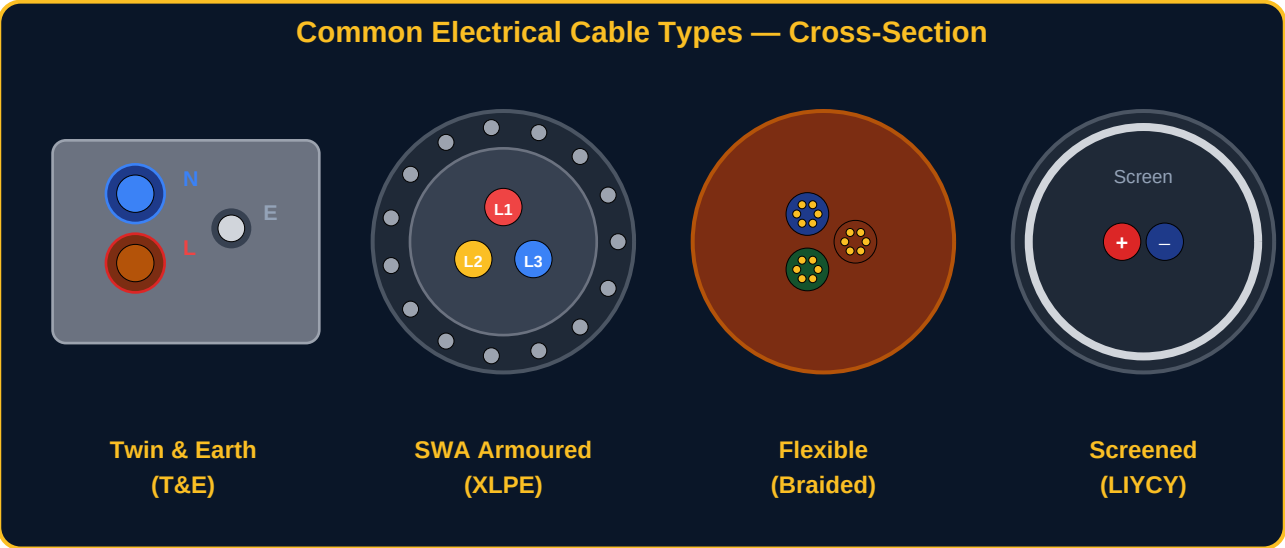
Electrical wiring is the physical backbone of every electrical installation — it connects sources of power to loads safely, reliably, and in compliance with regulations. Understanding cable construction, selection criteria, installation methods, and connection techniques is essential for every electrical engineer, from building services to industrial control panels to renewable energy systems. This lesson covers the fundamentals that inform every wiring decision.

Learning Objectives

- Identify the major cable types and their construction, insulation materials, and applications
- Select conductor cross-section using current rating tables and voltage drop calculations
- Apply IEC 60364 / BS 7671 installation method reference letters to determine current ratings
- Recognise and use the correct wire colour codes for IEC, old UK, and USA standards
- Describe the main connection methods: terminal blocks, ferrules, crimp lugs, and butt connectors
- List common wiring mistakes and explain how to avoid them professionally

1. Common Cable Types

Cable selection begins by matching the cable construction to the operating environment, mechanical protection needs, voltage class, and installation method. The wrong cable type causes premature insulation failure, fire risk, or voltage drop problems.



Cable Type	Construction	Voltage Rating	Typical Applications
Twin & Earth (T&E;)	Flat PVC sheath, 2 PVC-insulated cores (live+neutral) + bare earth. Available 1mm² to 16mm².	300/500V	Domestic and light commercial wiring in walls, ceilings, under floors. NOT for outdoor or damp locations without further protection.

SWA — Steel Wire Armoured	Copper conductors + XLPE or PVC insulation + inner sheath + steel wire armour + outer PVC sheath. 3-core to 5-core.	0.6/1kV	Underground runs, outdoor cables, industrial power feeds, anywhere mechanical damage risk is high.
Flexible (PVC or rubber)	Finely stranded copper conductors, coloured PVC or rubber insulation, braided overall sheath. Very flexible.	300/500V	Appliance leads, extension cables, luminaire connections, portable tools. NOT for fixed wiring in walls.
LIYCY Screened	Braided copper screen over twisted pairs with individual PVC insulation. Screened to reject EMI.	300/500V	PLC control wiring, instrumentation signals (4–20 mA), sensors, encoders in industrial environments with inverter drives.
LSZH (Low Smoke Zero Halogen)	Similar to PVC cables but LSZH compound sheath. Does not emit toxic/corrosive gases in fire.	0.6/1kV	Schools, hospitals, tunnels, confined spaces, public buildings — anywhere fire smoke toxicity is a safety concern.

2. Conductor Sizing — Three Criteria

A conductor must be sized to satisfy three independent requirements. The **largest cross-section** from the three criteria governs the final selection. Undersizing even one criterion creates a fire hazard, regulation non-compliance, or equipment malfunction.

Criterion	What It Prevents	Governing Standard	How to Apply
Current-carrying capacity (CCC)	Overheating — conductor temperature must stay below insulation rating (70°C PVC, 90°C XLPE)	IEC 60364-5-52, BS 7671 Appendix 4	Select conductor size from CCC table for the installation method. Apply de-rating factors for temperature, grouping, and burial depth.
Voltage drop	Load equipment operating below minimum voltage — motors overheat, lighting flickers, electronics malfunction	IEC 60364-5-52, BS 7671 Appendix 12	Calculate $VD = (2\rho LI)/A$ for single-phase or $(\sqrt{3}\times\rho LI)/A$ for three-phase. VD must be $\leq 3\%$ for lighting, $\leq 5\%$ for power circuits.
Short-circuit protection	Cable insulation damage from fault current during the time taken for the protective device (fuse/MCB) to clear the fault	IEC 60364-4-43, IEC 60909	Use adiabatic equation: $A_{min} = (I_{fault} \times \sqrt{t}) / k$. k depends on conductor and insulation material (k=115 for Cu/PVC at 70°C).

Voltage Drop — Formula and Diagram

Voltage Drop in a Cable Run — Why It Matters

SOURCE
230V AC

Cable (length L, cross-section A, resistivity ρ)

LOAD
224V

Length = L (metres)

$$VD = (2 \times \rho \times L \times I) / A \rightarrow VD\% = (VD / V_{supply}) \times 100$$

IEC limit: $VD \leq 3\%$ for lighting, $\leq 5\%$ for power. BS 7671 limit: $\leq 3\%$ from origin.

Current Rating De-rating Factors

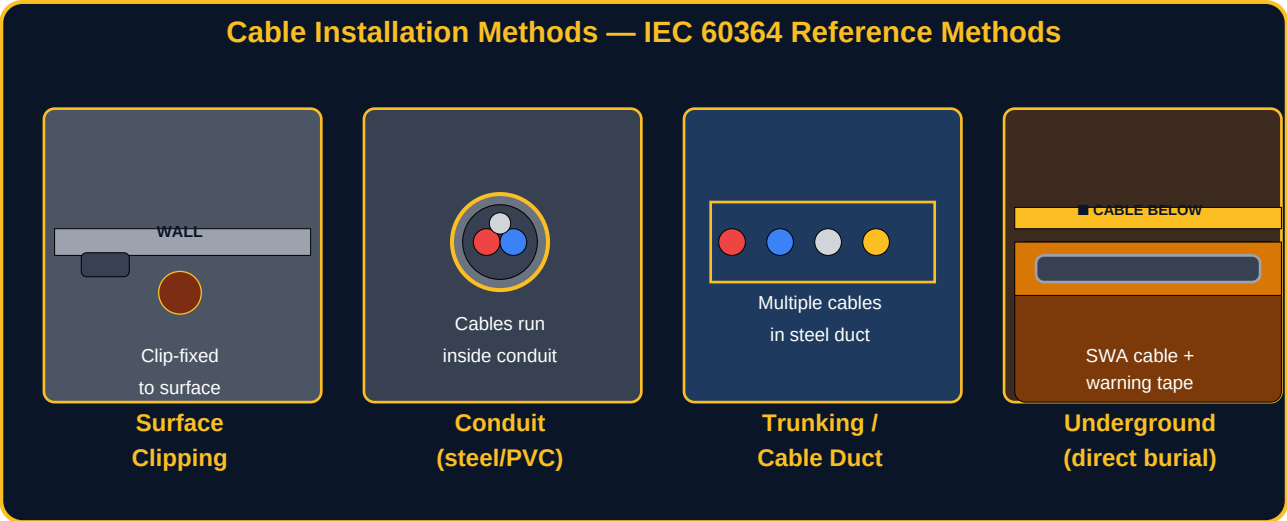
The tabulated current rating assumes ideal conditions. Real installations require multiplying the base rating by correction factors (Cf) to get the derated rating. If the derated rating is below the circuit design current, the next larger conductor size must be selected.

Factor	When Applied	Typical Value Range
Ambient temperature (Ca)	Air temperature different from 30°C (IEC) or 25°C (BS 7671)	$C_a = 0.77$ at 45°C; 0.91 at 35°C; 1.15 at 10°C (for 70°C PVC insulation)

Grouping (Cg)	Multiple cables clipped together or drawn through same conduit	$C_g = 0.80$ for 3 circuits, 0.70 for 6 circuits, 0.55 for 12 circuits (touching)
Thermal insulation (Ci)	Cable surrounded by thermal insulation (e.g. in ceiling void with rockwool)	$C_i = 0.50$ for fully surrounded. Install cables BELOW insulation wherever possible.
Buried depth (Cd)	Direct-buried cables at different depths from 0.5m standard	Typically 0.96 – 1.04 for shallow vs deep burial
Derated rating I_z	Combined all factors	$I_z = I_{tab} \times C_a \times C_g \times C_i \times C_d \geq I_b$ (design current)

3. Cable Installation Methods

The installation method directly affects the cable's current-carrying capacity — cables buried in walls or grouped in conduit dissipate heat less efficiently than cables clipped in open air. IEC 60364 and BS 7671 define reference letters (A1, A2, B1, B2, C, D1, D2, E, F) for each method, each with its own current rating table.



IEC Ref.	Installation Method	Heat Dissipation	Current Rating vs Method C
A1	In conduit in thermally insulated wall	Very poor — surrounded by insulation	50–60% of open-air clip rating
A2	Multi-core cable in conduit in insulated wall	Very poor	55–65% of clip rating
B1	In conduit on wall or in trunking on wall	Moderate	70–80% of clip rating
B2	Multi-core in trunking	Moderate	75–85% of clip rating
C	Clipped direct to surface (open air)	Good — free air on most surfaces	100% (reference method)
E/F	Free air — suspended cable or on cable tray	Excellent — air circulation all sides	105–120% of clip rating
D1	Underground in conduit, 0.7m depth	Good — ground acts as heat sink	90–100% of clip rating
D2	Underground direct buried, 0.7m depth	Excellent — direct contact with soil	100–115% of clip rating

4. Wire Colour Codes

Wire colour coding prevents dangerous misidentification of conductors. Two major standards are in use worldwide. Engineers working on imported equipment or in different countries **must verify the colour standard in use** before connecting any circuit — the same colour means different things in different standards.

Conductor	IEC 60446 / EU / New UK (post 2004)	Old UK BS 7671 (pre 2004)	USA NEC / NFPA 70	Note
Phase L1	Brown	Red	Black	CRITICAL: Brown (IEC) = Black (USA)
Phase L2	Black	Yellow	Red	Black = Neutral in old UK!
Phase L3	Grey	Blue	Blue	
Neutral (N)	Blue	Black	White or Grey	IEC Blue = USA White. Old UK Black = USA Black (L1)!
Earth / PE	Green/Yellow striped	Green/Yellow striped	Green or Green/Yellow	Universal — always green/yellow
DC Positive (+)	Red (recommended)	Red	Red	No mandatory standard for DC in many countries
DC Negative (-)	Blue or Black	Black	Black	Verify at every installation — no global DC colour standard
Control (AC)	Red	Red	Red	24VAC or 110VAC control wiring, not to be confused with L1

5. Electrical Connection Methods

Every conductor termination in a professional electrical installation uses a proper mechanical connection method. Twisted conductors pushed into a hole, tape-wrapped joints, and loose contacts are all causes of fire, equipment damage, and regulation non-compliance.

Electrical Connection Methods

Screw Terminal Block



Wire clamped by screw pressure
Direct contact
BS EN 60947-7

Ferrule / End Sleeve



Prevents strand scatter in terminals
Required on all
Mandatory >16 mm²

Crimp Lug (Ring/Spade)



Bolted to busbar or equipment stud
Ratchet crimp only
IEC 62291, DIN 46234

Butt Connector



Two wires crimped end-to-end
Same cross-section
Not reusable

Connection Type	When to Use	Standards	Never Do This
Screw terminal block (DIN rail)	All control panel and distribution board wiring. Stranded wire must have ferrule fitted first before inserting into screw terminal.	IEC 60947-7-1, EN 60947	Never insert bare stranded wire directly — strands loosen over time and cause arcing
Ferrule / end sleeve	On ALL stranded wire ends before screw terminals — mandatory above 6mm². Ensures all strands are captured.	DIN 46228	Never skip ferrules on fine-stranded wires — one loose strand touching an adjacent terminal = short circuit
Crimp ring/spade lug	Connecting cables to busbar bolts, motor terminals, equipment studs. Hydraulic or ratchet crimp tool required.	IEC 62291, BS EN 46234	Never use a hammer or pliers to crimp lugs — only proper ratchet crimp tool ensures correct compression
Butt connector	Joining two cables in conduit where a full terminal block is not accessible. Use heat-shrink type.	IEC 60947-7	Never use non-insulated butt connectors in damp or outdoor locations
Junction box / enclosure	All junction and termination points must be in an accessible enclosed box. No buried or hidden joints.	IEC 60364-5-52	Never bury cable joints underground without an accessible inspection chamber — violates all wiring regulations

6. Wiring Regulations Overview

Electrical wiring must comply with national and international standards. All regulations share the same goal: preventing electric shock, fire, and damage to equipment. The engineer's

responsibility is to know which standard applies to the project location and client requirements.

Standard	Region / Country	Scope	Key Sections Relevant to Wiring
IEC 60364	International (base standard)	Complete LV installation design, protection, wiring, testing	60364-5-52 (cable selection), 60364-4-43 (overcurrent protection), 60364-6 (testing)
BS 7671 (18th Ed.)	United Kingdom + many former Commonwealth nations	UK national adoption of IEC 60364 with UK-specific amendments	Appendix 4 (CCC tables), Appendix 12 (voltage drop), Part 6 (inspection & testing)
NFPA 70 (NEC)	United States of America	Electrical installations in buildings and industrial premises	Articles 310 (conductors), 300 (methods), 230 (services), 250 (grounding)
SANS 10142	South Africa	Based on IEC 60364 with local amendments	Volume 1: low-voltage installations, Volume 2: high-voltage
NFC 15-100	France	French national standard for LV installations	Based on IEC 60364; widely used in French-speaking Africa

7. Common Wiring Mistakes and How to Avoid Them

Mistake	Risk	Correct Practice
Undersized conductor (based only on current rating, ignoring voltage drop)	Equipment voltage too low — motors overheat, lights flicker, VFD faults on undervoltage	Always check all three criteria: CCC, voltage drop, and fault protection. Select largest of three results.
Using PVC cable in high-temperature location (near boiler, in engine bay)	PVC softens above 70°C → insulation fails → short circuit and fire	Use XLPE (rated 90°C) or silicone-insulated cable (rated 180°C) in hot areas
Bare stranded wire in screw terminal without ferrule	Strand loosens over time, migrates to adjacent terminal → short circuit, arcing, fire	Fit correct ferrule to all stranded conductors before inserting into screw terminals
Cable runs mixed with different voltage/signal levels in same trunking	Induced noise on signal cables → control system faults, false trips, PLC I/O errors	Separate power cables (50mm min) and instrument cables (150mm min) in dedicated trunking
Reversed neutral and earth at socket outlet	Appliance chassis becomes live — electric shock hazard even with appliance switched off	Test every outlet with a socket tester after installation. Earth must be 0V to true earth.
No cable support — cable hanging freely over long span	Gravity stress on terminations → loosening at screw terminals over time	Support every 250mm for 1.5mm ² T&E; 400mm for 4mm ² , 550mm for 10mm ² . Use proper cable cleats.
Armoured cable gland without proper earth bonding	SWA armour not earthed → armour becomes live in fault → electric shock, cable fire not cleared by protection	Use CW glands with lock-nuts. Bond armour to earth bar via earth tag on gland.

8. Practice Exercises

#	Exercise
1	Cable selection: A 3-phase 400V motor (22 kW, FLA=42A, PF=0.85) is fed from an MCC 35m away. Cables are clipped on cable tray (Method C). Ambient 35°C. Select cable size satisfying CCC. Then check: is voltage drop within 5%? Use $k=115$ for Cu/PVC.
2	Voltage drop: A 32A lighting circuit runs 28m from a consumer unit. Single-phase 230V. Conductor is 4mm ² Cu. Calculate voltage drop in volts and as %. Does it comply with the 3% limit? ($\rho_{Cu} = 1.72 \times 10^{-8} \Omega \cdot m$)
3	De-rating: A 16mm ² cable rated 87A (Method C) passes through a loft filled with 150mm fibreglass insulation (fully surrounded: $C_i=0.5$) at an ambient temperature of 40°C ($C_a=0.87$). What is the derated current rating? Can it carry a 35A load?
4	Colour identification: You are working on a 3-phase DB installed in 2000 (old UK colours). The phase wires are Red, Yellow, Blue. The client wants to add a new circuit using IEC-harmonised cable. How do you correctly identify and label the existing wiring before extending it?
5	Installation method: An industrial machine requires power and control cables from a panel to a remote junction box 20m away. Power cable is 3-core 4mm ² SWA. Control cables are two LIYCY 4×0.75mm ² screened. Specify the installation method and explain cable separation requirements.

Lesson Summary

Topic	Key Rule
Cable selection	Match cable type to environment first (T&E; indoors, SWA outdoors/underground, screened for signals).
Conductor sizing — three checks	Size for: (1) CCC with de-rating factors, (2) voltage drop $\leq 3\text{--}5\%$, (3) fault protection adiabatic. Use LARGEST result.
De-rating	Multiply tabulated CCC by all applicable factors: Ca (temp), Cg (grouping), Ci (insulation). Always apply ALL relevant factors simultaneously.
Voltage drop limit	IEC 60364 / BS 7671: $\leq 3\%$ lighting, $\leq 5\%$ power. $VD = (2pLI)/A$ (single-phase) or $(\sqrt{3}pLI)/A$ (three-phase).
Installation method	Method C (clipped in air) = 100% reference. Underground or grouped in conduit = derated. Reference method determines which column to use in IEC 60364-5-52 Table B.52's.
Wire colours	ALWAYS verify which standard applies: IEC/EU (Brown-Black-Grey-Blue-GY) vs USA NEC (Black-Red-Blue-White-GY). NEVER assume.
Connections	Ferrule on every stranded end. Proper crimp tool for lugs. No bare joints outside a junction box. Earth bond SWA armour at both ends.
Wiring regulations	Know which standard applies by project location. IEC 60364 = international. BS 7671 = UK. NEC = USA. All require inspection certificate on completion.

What Is Next — ELEC-008

ELEC-008: Electrical Symbols — Complete reference for IEC 60617 and ANSI/IEEE symbol standards. Switches, relays, contactors, motors, transformers, protection devices, luminaires, and instrumentation symbols. How to read and draw single-line diagrams, schematic diagrams, and panel wiring diagrams using standard symbol libraries.

Resource	Details
PDF Series	ELEC-001 through ELEC-030 at ayetechub.com/pdfs.html
Wiring Diagrams eBook	Wiring Diagram Mastery — free flipbook at ayetechub.com
Community	AYE Tech Hub Telegram: t.me/ayetechub